

SpacemiT

K3 Pico-ITX

RISC-V Architecture For Intelligent Future



Ultra-High Integration Mini AI Computer

The K3 Pico-ITX is a single-board computer delivering up to 60 TOPS of AI performance. It features a unified memory architecture with 8 CPU cores and 8 AI acceleration cores, along with onboard high-speed UFS storage and a 10Gb optical networking interface. This design maximizes computing capability and improves efficiency for applications such as scientific computing and AI.

Built in a 2.5-inch Pico-ITX Plus form factor, the K3 Pico-ITX is optimized for space-constrained deployments. The board supports dual M.2 expansion slots and provides interfaces for real-time motion control and system management. With rich I/O expandability and an industrial-grade architecture, the K3 Pico-ITX enables rapid evaluation and system integration, accelerating time to market.



Unified AI Compute

- 8-core K3 processor compliant with the RVA23 profile, delivering 60 TOPS of AI performance with IME extension support and full virtualization
- Unified memory architecture across compute and AI cores, enabling deployment of 30B models



Ready Out of the Box

- Onboard UFS storage, offering speeds up to 3.4× faster than typical eMMC solutions
- Full-featured USB Type-C with 65W PD and 4K DP—power and display in one cable



Rapid Integration

- Onboard eDP interface for HD displays, simplifying display system integration
- Flexible expansion I/O powered by the RT24 real-time core, supporting EtherCAT, CAN-FD, and other interfaces for microsecond-level motion control and robotics
- MUSE architecture with thermal and workload partitioning, ensuring optimal CPU performance and efficiency



Flexible Expansion

- Dual M.2 slots (B-Key and M-Key), with M-Key supporting 4-lane PCIe Gen3 for full-bandwidth NVMe SSD expansion
- Integrated 10GbE over PCIe with support for 10GBASE-R optical interface, enabling low-latency, high-throughput data transfer and scalable clustering



Specifications

Processor	SpacemiT K3, 8 cores @2.4 GHz, 60 TOPS AI performance, RVA23 compliant, supports IME vector extensions and full virtualization
Display	DP Type-C: up to 4K (3840×2160) @ 60 Hz 40-pin eDP: up to 2.5K (2560×1600) @ 90 Hz
Memory	Dual-channel 2 × 32-bit LPDDR5 6400 MT/s, 16 GB / 32 GB options
Local Storage	UFS 2.2, 128 GB / 256 GB options
Storage Expansion	M.2 M-Key (PCIe Gen3 ×4), supports 2280 NVMe SSD
HS Expansion	M.2 B-Key (PCIe Gen3 ×2 + USB), supports 2242/3042 cards
Real-Time Expansion	FPC connector for EtherCAT, 5× CAN-FD, SPI, I ² C, UART, etc.
Wireless	Onboard Wi-Fi 6 + BT 5.2, dual-band, dual-antenna, 802.11 a/b/g/n/ac/ax compliant
Wired Network	1× RJ45 Ethernet, 10/100/1000 Mbps adaptive
Optical Network	10GbE SFP+ port, supports 10GBASE-R / 10GBASE-X, QinQ, MSI-X, WOL, and clustering
Audio	Onboard CODEC, internal audio input/output
USB	2× USB 3.2 Gen1 Type-C (1 full-featured, one OTG) 4× USB 2.0 Type-A Host
Debug	UART, JTAG, and 3 side buttons for power, reset, and firmware update
System Management	Onboard EC controller for power, thermal, and system status; includes I ² C/UART/GPIO expansion
Form Factor	100 × 86 mm, Pico-ITX Plus single-board computer, approx. size of a 2.5" drive
OS	Pre-installed Bianbu 3.0; supports Ubuntu 26.04, OpenHarmony 6.0, OpenKylin, Deepin, Fedora, etc.
Power Input	Type-C USB-PD (65 W) or ATX 2-pin 12 V @ 7 A
Reliability	ESD protection: board: ±4 kV (contact), ±8 kV (air); system: ±6 kV (contact), ±12 kV (air) Compliant with CCC, CE, and FCC; operating temp: -20 °C ~ 70 °C (consumer) / -40 °C ~ 85 °C (industrial)
Clock	Onboard RTC with battery interface, supports G3 state
Structure	Optional single-board or fan-cooled heatsink assembly Optional single-board or fully-metal industrial chassis Optional real-time board, touchscreen, terminal blocks

Note: The M.2 B-Key PCIe ×2 lanes are shared with the M-Key slot; when both are populated, the M.2 M-Key operates at PCIe Gen3 ×2.

Optional Components

Category	Name	Description	Interface
Peripheral	SSD	980 NVMe™ M.2 SSD, PCIe Gen 3.0 ×4,NVMe 1.4, sequential read up to 3,500 MB/s, sequential write up to 3,000 MB/s	M.2 M-KEY 2280
	SSD	B+M NVMe SSD 128 GB	M.2 B-KEY 2242
	4G Module	EM05	M.2 B-KEY 3042
	SATA Expansion Card	PCIe to 5 × SATA interface	M.2 M-KEY 2280
	Docking Station	Type-C dock for HD 4K display, PD charging, USB 3.0 for tablets/laptops (3-in-1)	Full-featured Type-C
	Ultra HD Display	16" 2.5K LCD, 90 Hz, 2560 × 1600	40-pin eDP
	Real-time Control Expansion Board	19 V input, 5 × CAN-FD, EtherCAT, RS232 & RS485 I/O, industrial-grade isolation protection	FPC
Structural Accessory	Embedded Fan Heatsink	Custom aluminum design	FAN
	Metal Chassis	120 × 120 × 48 mm self-developed metal chassis	BTN

Block Diagram

